Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/05/2023
PI Name:	Plawsky, Joel Sc.D.		
Project Title:	LOOP-based Constrained Vapor Bubbles		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Troy	**State:**	NY
Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017 Physical Sciences NNH17ZTT001N-17PSI-E: Use of the NASA Physical Sciences Informatics System – Appendix E
Start Date:	11/01/2019	**End Date:**	03/31/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2023 per NSSC information (Ed., 11/3/22) NOTE: End date changed to 10/31/2022 per PI; original end date was 10/31/2021 (Ed., 8/17/21)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	80NSSC20K0120		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Loop heat pipes are quite common heat transfer devices in satellites and space systems but often fail and the causes are not always readily apparent. We had built a prototype loop heat pipe as part of the original Constrained Vapor Bubble (CVB) program but only used it to look at refrigerants and especially to help understand meniscus stability. In the proposal, we would like to resurrect that system, based on a wickless, square cross-section CVB and use it to collect ground-based data that we can compare with the linear heat pipe CVB data obtained during CVB1 and CVB2 missions. In particular, we are concerned with mapping the vapor liquid interface, driving the system to the point where Marangoni flows become important, and understanding the mechanisms of failure or instability in wickless loop systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work will enhance the development of heat pipes which are not only used for space applications but also for cooling high heat generation devices on Earth. We are also augmenting and enhancing our interferometry techniques which will greatly expand the types of fluids and surfaces that we can investigate . This will have great utility in looking at the fundamental mechanisms governing phase change heat and mass transfer. Since phase change is a key process in distillation, power generation, refrigeration, and electronics cooling, the more we can learn and uncover the more efficient devices we will be able to manufacture. We have also expanded the scope of the project looking into simulations of evaporation and condensation on surface and molecular dynamics simulations of surface tension of fluid mixtures. The work will help expand our knowledge of how fluids evaporate on surfaces, why menisci oscillate during evaporation, and why mixtures of liquid may make for more efficient heat pipes than pure fluids alone. In addition the simulations may also point to the intermolecular mechanisms by which self rewetting fluids operate.
Task Progress:	During the course of this project we have built and instrumented a loop Constrained Vapor Bubble (CVB) system. We re-analyzed boiling data from CVB-1 to develop a new correlation between the Nusselt and Ohnesorge numbers. We looked at a new way to perturb the evolution equations for thin evaporating films and showed the influence of fluid type and substrate type on the evaporation rate. The new analyses also confirmed an earlier prediction that oscillations in the thin film serve to enhance the rate of evaporation. The oscillations also maximize heat transfer when they occur at a specific region of film thickness and not when they occur in the thin, adsorbed film itself. We developed a new capillary feeder system that allows us to investigate the effects of the Hamaker constant on evaporation by using atomic layer deposition to deposit films of varying refractive index. This led to the need to develop new procedures to analyze the data, as the old methods could not deal with bare surfaces that may have very low reflectivity. We developed a maze-based method to develop simplified representations of porous media that would allow us to build new wicking structures for heat transfer as well as separations equipment. Finally, we used molecular dynamics (MD) simulations to try to understand the trends exhibited by Surface Light Scattering Spectrometry measurements of the surface tension of pentane/isohexane mixtures. The work indicates the need to modify the force fields used in MD simulations as they tend to predict higher values of liquid density, which leads to higher values of interfacial tension.
Bibliography Type:	Description: (Last Updated: 10/14/2025)
Articles in Peer-reviewed Journals	Plawsky J. "Constrained Vapor Bubble Experiment (CVB) in the Light Microscopy Module (LMM)." Gravitational and Space Research. 2024 Apr;12(1):60-3. https://doi.org/10.2478/gsr-2024-0004 , Apr-2024
Articles in Peer-reviewed Journals	Rishty A, Lucas J, Croce J, Peach S, Hall N, Nahra H, Balasubramaniam R, Woodcock C, Khusid B, Plawsky J. "The effects of microgravity on mini-channel flow boiling and CHF behavior." International Journal of Heat and Mass Transfer. 2026 Jan 1;254:127598. https://doi.org/10.1016/j.ijheatmasstransfer.2025.127598 , Jan-2026
Dissertations and Theses	Yu, Jiaheng. "The effect of interfacial forces on the performance of a wickless heat pipe." Dissertation, Rensselaer Polytechnic Institute, November 2022. , Nov-2022
Dissertations and Theses	Pawar, A. "Interfacial phenomena in heat pipes: Microscopic and molecular dynamics study." Dissertation, Rensselaer Polytechnic Institute, May 2023. , May-2023